A Closer Look at GRUB, the Linux Bootloader

Acquaint yourself with the first program that runs when you switch on your Linux computer. This is the program responsible for loading and transferring control to the OS kernel software. GRUB is an acronym for Grand Unified Bootloader.

Have you ever wondered which program runs first whenever a computer is switched on? Well, the answer is the bootloader. It primarily manages and completes the boot sequence of a computer system. It procures the OS kernel from the hard disk or any specified boot device within the boot sequence, into the main memory.

GRUB (Grand Unified Bootloader) is the default bootloader for many Linux distributions. Apart from GRUB, the other most common bootloaders for Linux are LILO (LInux LOader) and LOADLIN (LOAD).

GRUB is a favoured bootloader as it is free and unrestricted software. It has also got an incredible graphical interface, where a user can select the preferred operating system from the menu. The GUI also works in exactly the same way as a command line interface. The difference is that a pre-configured configuration file contains the command which identifies the hard disk and kernel location (which have to be typed by hand, if you are using the command line interface).

The following site will be useful for the GRUB commands that will help to troubleshoot most of the booting problems: *https://access.redhat.com/documentation/en-US/Red_Hat_Enterprise_Linux/3/html/Reference_Guide/s1-grub-commands.html*

GRUB is most often used on UNIX-like systems including GNU, Linux and Solaris. The new version of GRUB that we now use is GRUB2. New developments are continuing to take place, but the current free versions are quite functional for customary operations. The previous version of GRUB was known as GRUB Legacy.

GRUB2 is an upgraded version of GRUB Legacy, with the following enhancements:

- GRUB2 is proficient enough to govern a hard disk partition using UUIDs (Universally Unique Identifiers).
- The new GRUB2 configuration file is highly configurable, compared to the GRUB-Legacy config file. The new configuration file is called *grub.cfg*. Earlier, we used *menu.lst* or *grub.config.*
- GRUB2 also supports EFI (Extensible Firmware Interface), which is a specification that defines a software interface between an operating system and a platform firmware.
- GRUB2 recognises many more new file systems like ext-4.

In addition, GRUB2 has the following features:

- It is stable.
- Users can make changes during boot-up, hence GRUB is dynamically configurable.
- It supports an unlimited number of boot entries.
- GRUB can install to and run from any device including hard drives, floppy disks, DVDs, CD-ROMs and USB drives.
- The command interface is interactive.
- GRUB does not require the exact physical location of the operating system kernel in the hard disk. It just requires the hard disk number (like the first hard disk, second hard disk, etc), and the partition number, along with the filename of the kernel. GRUB understands the format in which the kernel is made.
- Booting options like kernel parameters can be modified through the command line.

To determine the version of GRUB that's on your system, use the following command:

```
grub-install -V
```

The GRUB bootloader presents users with a choice of kernels, and you can reboot into a known good kernel. Some distros (like Ubuntu) use a default GRUB configuration that hides that menu. You can get the menu to appear by pressing the *Esc* key during boot up after the BIOS display disappears.

To make the GRUB boot menu always appear under Ubuntu, run the following command:

```
sudo vim /etc/default/grub
```

We can also set the time for GRUB to show the menu before selecting the kernel at the top of the list, by typing the following in the terminal:

```
GRUB_TIMEOUT=20
```

Here, I have set the time interval as 20 seconds. After making changes in GRUB, we have to update it. So run the following command:

```
sudo update-grub2
```

Then, reboot your system.

GNU GRUB is rich with capabilities beyond what are described here. For more information, you can check the GNU GRUB information manual given at *https://www.gnu.org/software/grub/manual/grub.html*. END

References

[1] *http://www.gnu.org*
[2] *http://searchenterpriselinux.techtarget.com/definition/GNU-GRUB*

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